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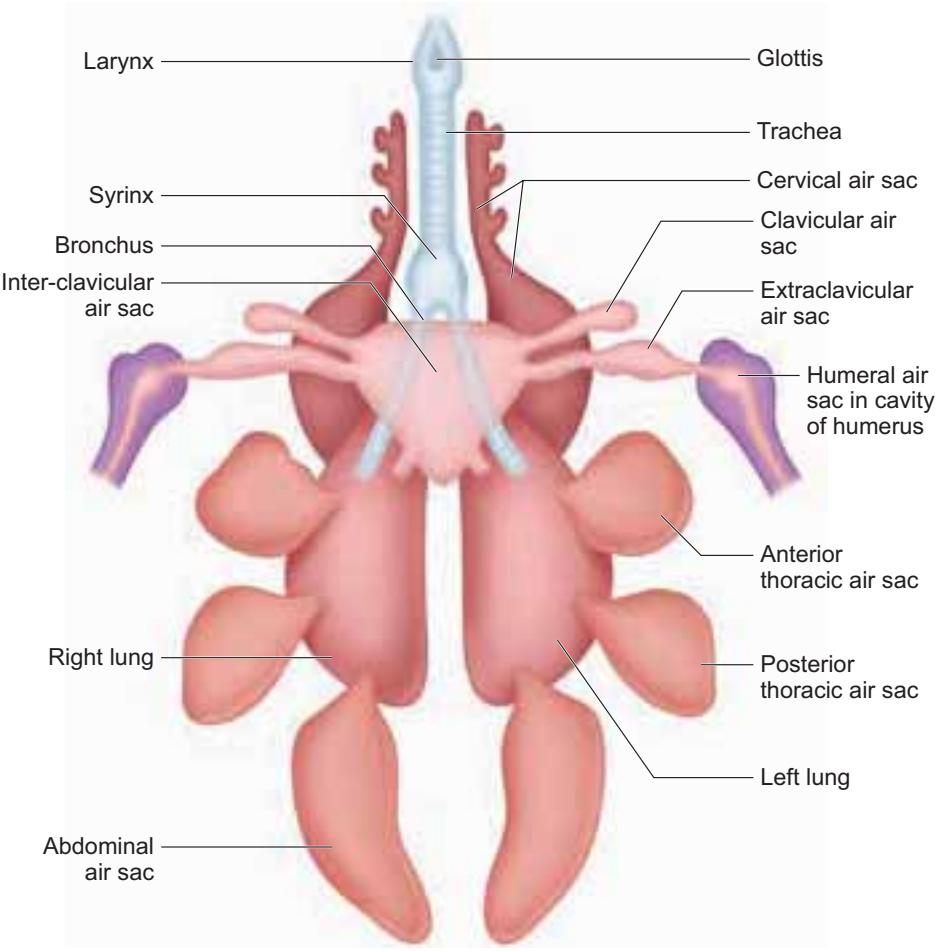

Figure 4.30 Pigeon – Respiratory System in ventral view

mixed with the **bile** and **pancreatic juice**, and further digestion is effected.

Respiratory system

In birds the type of respiration is **pulmonary.** The respiratory system includes the respiratory tract, the respiratory organs and air sacs. A true muscular diaphragm is absent in birds. The **respiratory tract** includes the nares, nasal sacs, glottis, larynx, trachea and syrinx. The **respiratory organs** are the lungs and air sacs. The larynx opens into the trachea and is supported by a series of closely set rings. The trachea divides into two **bronchi**, each of which divides and sub-divides into smaller branches, ultimately ending in fine air-capillaries which lies intermingled with the capillaries of the pulmonary vessels. Lungs are solid spongy organs; attached dorsally to the ribs. There are nine air-sacs: a pair of **cervical sacs** at the base of the neck one on each side; a single median **interclavicular air sac** connected with both lungs and situated in between the two limbs of the furcula and on either sides it gives off an **extraclavicular air sac** communicating with an air - cavity of the humerus and a **clavicular air sac**; two pairs of **thoracic air sacs** and a pair of **abdominal air sacs**. This complicated arrangement adds to the efficient respiratory function and maintenance of a high temperature (Figure 4.30).

Respiratory mechanism

The lungs are not dilatable since the skeleton around them forms a rigid framework. Inspiration is passive and expiration is an active process. During respiration the sternum